

BOGORODITSKOE-FENINO, Russian Federation

WMO: 341100

Lat: **51.167N**Lon: **37.350E**Elev: **740**StdP: **14.31**Time Zone: **3.00 (E03)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-3.3	-13.6	2.7	-7.8	-8.9	3.5	-2.9	17.6	20.5	16.4	21.7	4.8	50	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	89.6	68.2	86.0	66.8	82.8	66.0	70.4	83.3	69.1	81.6	67.8	79.7	6.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	99.1	75.6	64.9	94.6	73.6	63.8	91.0	72.5	34.7	83.5	33.6	81.5	32.6	80.0	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
16.6	14.7	13.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-13.0	94.4	7.2	3.9	-18.2	97.2	-22.4	99.5	-26.4	101.7	-31.6	104.6		
			-13.4	73.1	7.0	1.2	-18.5	74.0	-22.6	74.7	-26.5	75.4	-31.6	76.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	18.3	22.1	30.9	47.6	60.6	66.2	69.5	67.5	57.4	44.4	34.8	26.5
	DBStd	19.81	11.57	11.40	8.71	8.38	6.60	6.47	5.91	6.88	7.19	8.38	8.59	10.10
	HDD50	3928	981	782	592	145	6	0	0	0	19	216	458	729
	HDD65	7506	1446	1202	1056	523	163	65	25	50	239	638	905	1194
	CDD50	2329	0	0	0	74	336	486	604	543	242	42	2	0
	CDD65	436	0	0	0	1	28	101	164	129	13	0	0	0
	CDH74	4407	0	0	0	23	335	890	1484	1458	217	0	0	0
	CDH80	1394	0	0	0	2	55	218	516	573	30	0	0	0
Wind	WSAvg	6.3	7.0	7.1	7.2	6.9	5.7	5.4	5.0	5.0	5.6	6.2	6.6	7.5
Precipitation	PrecAvg	22.0	1.5	1.2	1.2	1.6	1.9	2.5	2.6	2.4	2.0	1.7	1.9	1.7
	PrecMax	31.2	3.6	2.2	2.8	4.2	4.7	5.4	5.0	4.9	4.4	3.4	5.0	4.3
	PrecMin	14.5	0.2	0.1	0.3	0.2	0.2	0.6	0.5	0.3	0.2	0.0	0.2	0.1
	PrecStd	3.9	0.9	0.5	0.6	1.0	1.1	1.3	1.1	1.3	1.2	1.0	1.2	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	40.3	58.7	76.6	85.0	88.9	95.0	97.4	83.7	69.5	56.9	46.4
		MCWB	37.3	38.4	47.4	57.0	63.7	67.6	68.6	67.7	63.2	57.3	52.7	44.7
	2%	DB	36.4	37.9	49.7	72.2	81.1	85.4	89.6	90.5	79.5	65.3	53.6	43.3
		MCWB	34.8	36.0	42.4	54.5	62.0	66.7	68.6	68.0	61.4	55.3	50.9	42.1
	5%	DB	34.6	35.8	45.1	68.0	77.9	82.3	85.7	86.2	76.3	60.3	49.8	40.4
		MCWB	33.3	34.1	39.8	53.1	61.5	66.8	68.0	66.8	60.1	53.6	47.2	39.2
	10%	DB	32.7	34.3	41.4	63.0	74.3	79.3	81.9	82.3	71.7	56.7	45.9	37.9
		MCWB	31.6	33.0	37.4	50.4	60.3	65.5	66.7	65.5	58.7	52.0	44.1	36.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	39.0	47.6	59.4	67.0	71.3	72.5	71.7	67.0	59.6	53.6	45.0
		MCDB	38.6	39.8	56.0	74.6	78.4	82.4	88.1	86.4	77.8	65.9	55.8	45.9
	2%	WB	35.3	36.7	43.2	55.7	65.2	69.7	70.6	69.6	63.2	57.0	51.0	42.3
		MCDB	36.1	37.7	48.1	69.1	76.5	81.2	83.7	85.4	75.8	62.6	53.6	43.3
	5%	WB	33.4	34.6	40.2	53.5	63.6	68.3	69.3	68.1	61.2	54.7	47.4	39.5
		MCDB	34.3	35.3	44.2	64.6	74.0	79.3	81.2	83.2	72.2	59.7	49.5	40.3
	10%	WB	31.7	33.1	38.0	51.5	61.7	66.8	68.1	66.5	59.2	52.0	44.5	36.8
		MCDB	32.7	34.0	41.2	62.0	70.9	76.5	79.2	79.7	70.3	56.1	46.0	37.8
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.3	11.9	18.4	20.0	19.4	19.2	21.5	18.7	13.7	8.4	7.3
		MCDBR	6.6	7.4	17.7	24.9	25.3	23.7	25.3	27.6	27.7	20.3	12.6	7.9
		MCWBR	6.4	6.5	11.4	12.1	11.0	10.1	9.0	10.1	12.5	11.9	10.5	7.7
	5% WB	MCDBR	6.3	6.5	16.1	22.5	20.6	20.3	20.9	24.6	23.6	18.1	12.1	7.7
		MCWBR	6.2	6.2	11.2	11.7	10.2	9.5	8.7	10.0	12.4	11.9	10.4	7.7
Clear-Sky Solar Irradiance	taub	0.305	0.338	0.367	0.446	0.401	0.390	0.424	0.430	0.392	0.361	0.338	0.308	
	taud	2.124	2.041	2.104	2.071	2.291	2.341	2.257	2.224	2.303	2.348	2.281	2.186	
	Ebn at Noon	215	239	259	250	271	275	262	253	250	235	205	194	
	Edh at Noon	24	35	40	47	39	38	41	40	33	26	22	20	
All-Sky Solar Radiation	RadAvg	303	617	1031	1321	1812	1917	1843	1625	1073	578	255	183	
	RadStd	63	91	166	125	140	160	130	140	137	89	53	41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (6 neighbors)	+1.48	N/A	N/A	N/A	N/A	N/A	-283	-407	+220	+90